POISSON

$$f(\mu_i) = \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!} \qquad \mu_i = e^{\tilde{x}_i^T\beta}$$

LIKELIHOOD

$$L(\beta) = \prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}$$

LOG-LIKELIHOOD

$$\lambda(\beta) = \log(L(\beta)) = \log\left(\prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum \log(e^{-\mu_i}) + \log(\mu_i)^{y_i} - \log(y_i!) = \sum -\mu_i + y_i\log(\mu_i) - \log(y_i!) = \sum -e^{\tilde{x}_i^T\beta} + y_i\tilde{x}_i^T\beta - \log(y_i!)$$

SCORE FUNCTION

$$\ell_*(\beta) = \begin{cases} a = \frac{d}{d\beta_1}\lambda(\beta) \\ b = \frac{d}{d\beta_2}\lambda(\beta) \end{cases}$$

$$a = \frac{d}{d\beta_1}\left[\sum -e^{\tilde{x}_i^T\beta} + y_i\tilde{x}_i^T\beta - \cancel{\log(y_i!)}\right] = \frac{d}{d\beta_1}\left[\sum -e^{\beta_1+\beta_2x_i} + y_i(\beta_1+\beta_2x_i)\right] = \sum -e^{\beta_1+\beta_2x_i} + y_i$$

$$b = \frac{d}{d\beta_2}\left[\sum -e^{\tilde{x}_i^T\beta} + y_i\tilde{x}_i^T\beta - \cancel{\log(y_i!)}\right] = \sum -e^{\beta_1+\beta_2x_i}\cdot x_i + y_ix_i$$

$$\hat{\beta}(Y) \dot{\sim} N_p(\beta;\ J(\hat{\beta})^{-1})$$

MARGINAL $\quad \hat{\beta}_r(Y) \dot{\sim} N(\beta_r;\ J(\hat{\beta})^{-1}_{(r,r)})$

BERNOULLI

$$f(\pi_i) = \pi_i^{y_i}(1-\pi_i)^{1-y_i} \qquad \pi_i = \frac{e^{\tilde{x}_i^T\beta}}{1+e^{\tilde{x}_i^T\beta}}$$

LIKELIHOOD

$$L(\theta) = \prod \pi_i^{y_i}(1-\pi_i)^{1-y_i}$$

LOG-LIKELIHOOD

$$\lambda(\theta) = \log(L(\theta)) = \log\left(\prod \pi_i^{y_i}(1-\pi_i)^{1-y_i}\right) = \sum \log(\pi_i^{y_i}) + \log(1-\pi_i)^{1-y_i} = \sum y_i\log(\pi_i) + (1-y_i)\log(1-\pi_i) = \sum y_i\log\left(\frac{e^{\tilde{x}_i^T\beta}}{1+e^{\tilde{x}_i^T\beta}}\right) + (1-y_i)\log\left(\frac{1}{1+e^{\tilde{x}_i^T\beta}}\right)$$

$$= \sum y_i\tilde{x}_i^T\beta - \cancel{y_i\log(1+e^{\tilde{x}_i^T\beta})} - (1-\cancel{y_i})\log(1+e^{\tilde{x}_i^T\beta}) = \sum y_i\tilde{x}_i^T\beta - \log(1+e^{\tilde{x}_i^T\beta})$$

SCORE FUNCTION

$$\ell_*(\theta) = \begin{cases} a = \frac{d}{d\beta_1}\lambda(\theta) \\ b = \frac{d}{d\beta_2}\lambda(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1}\left[\sum y_i\tilde{x}_i^T\beta - \log(1+e^{\tilde{x}_i^T\beta})\right] = \frac{d}{d\beta_1}\left[\sum y_i(\beta_1+\beta_2x_i) - \log(1+e^{\beta_1+\beta_2x_i})\right] = \sum y_i - \frac{e^{\beta_1+\beta_2x_i}}{1+e^{\beta_1+\beta_2x_i}}\cdot 1$$

$$b = \frac{d}{d\beta_2}\left[\sum y_i(\beta_1+\beta_2x_i) - \log(1+e^{\beta_1+\beta_2x_i})\right] = \sum y_ix_i - \frac{e^{\beta_1+\beta_2x_i}}{1+e^{\beta_1+\beta_2x_i}}\cdot x_i$$

DISTRIBUTION

$$\hat{\beta}(Y) \dot{\sim} N_p(\beta;\ J(\hat{\beta})^{-1})$$

MARGINAL $\quad \hat{\beta}_r(Y) \dot{\sim} N(\beta_r;\ J(\hat{\beta})^{-1}_{(r,r)})$

$$f(\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\} \qquad \mu_i = \beta_1 + \beta_2 x_i$$

LIKELIHOOD $L(\theta)$

$$L(\theta) = \prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}[y_i - \mu_i]^2\right\}$$

LOG-LIKELIHOOD

$$l(\theta) = \log(L(\theta)) = \log\left(\prod \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right\}\right) = \sum \log\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right\}\right) = \sum -\log(\sqrt{2\pi\sigma^2}) - \frac{1}{2\sigma^2}\sum(y_i - \mu_i)^2 = \sum -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(y_i - \mu_i)^2$$

$$= -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(y_i - \mu_i)^2 = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(y_i - \beta_1 - \beta_2 x_i)^2$$

SCORE FUNCTION

$$l_*(\theta) = \begin{cases} a = \frac{d}{d\beta_1} l(\theta) \\ b = \frac{d}{d\beta_2} l(\theta) \\ c = \frac{d}{d\sigma^2} l(\theta) \end{cases}$$

$$a = \frac{d}{d\beta_1} -\frac{1}{2\sigma^2}\sum(y_i - \beta_1 - \beta_2 x_i)^2 = -\frac{1}{2\sigma^2}\sum 2(y_i - \beta_1 - \beta_2 x_i)\cdot(-1) = \frac{1}{\sigma^2}\sum 2(y_i - \beta_1 - \beta_2 x_i)$$

$$b = \frac{d}{d\beta_2} -\frac{1}{2\sigma^2}\sum(y_i - \beta_1 - \beta_2 x_i)^2 = -\frac{1}{2\sigma^2}\sum 2(y_i - \beta_1 - \beta_2 x_i)\cdot(-x_i) = \frac{1}{\sigma^2}\sum(y_i - \beta_1 - \beta_2 x_i)(x_i)$$

$$c = \frac{d}{d\sigma^2}\left[\cancel{-\frac{n}{2}\log(2\pi)} - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\sum(y_i - \beta_1 - \beta_2 x_i)^2\right]$$

$$= -\frac{n}{2}\frac{1}{\sigma^2} - \frac{1}{2}\sum(y_i - \beta_1 - \beta_2 x_i)^2 \underbrace{\frac{d}{d\sigma^2}\frac{1}{\sigma^2}}_{\frac{d}{dx}\frac{1}{x} = \frac{d}{dx}x^{-1} = -x^{-2} = -\frac{1}{x^2}} = -\frac{n}{2}\frac{1}{\sigma^2} + \frac{1}{2}\frac{1}{(\sigma^2)^2}\sum(y_i - \beta_1 - \beta_2 x_i)^2$$

- $\hat{\underline{\beta}}(\underline{Y}) \dot\sim N_p(\underline{\beta}; \sigma^2(X^T X)^{-1})$
- $\hat{\beta}_r(Y) \dot\sim N(\beta_r; \sigma^2(X^T X)^{-1}_{(r,r)})$

POISSON

$$f(\mu_i) = \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!} \qquad \mu_i = e^{\beta_1+\beta_2 x_i}$$

LIKELIHOOD

$$L(\underline{\beta}) = \prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}$$

LOG-LIKELIHOOD

$$\ell(\underline{\beta}) = \log(L(\underline{\beta})) = \log\left(\prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum -\mu_i + y_i\log(\mu_i) - \log(y_i!) = \sum -e^{\beta_1+\beta_2 x_i} + y_i(\beta_1+\beta_2 x_i) - \log(y_i!)$$

SCORE FUNCTION

$$\ell_*(\underline{\beta}) = \begin{cases} \omega = \frac{d}{d\beta_1}\ell(\underline{\beta}) = \frac{d}{d\beta_1}\sum -e^{\beta_1+\beta_2 x_i} + y_i(\beta_1+\beta_2 x_i) - \log(y_i!) = \sum -e^{\beta_1+\beta_2 x_i}\cdot 1 + y_i \\ 6 = \frac{d}{d\beta_2}\ell(\underline{\beta}) = \sum -e^{\beta_1+\beta_2 x_i} + y_i(\beta_1+\beta_2 x_i) - \log(y_i!) = \sum -e^{\beta_1+\beta_2 x_i}\cdot x_i + y_i x_i \end{cases}$$

- $\hat{\underline{\beta}}(\underline{Y}) \dot\sim N_p(\underline{\beta}; J(\hat{\underline{\beta}})^{-1})$
- $\hat{\beta}_r(\underline{Y}) \dot\sim N(\beta_r; J(\hat{\underline{\beta}})^{-1}_{(r,r)})$

$$f(y_i;\theta)=\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right\}$$

LIKELIHOOD

$$L(\theta)=\prod\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right\}$$

$$=\prod(2\pi\sigma^2)^{-1/2}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right\}$$

$$=(2\pi\sigma^2)^{-n/2}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right\}$$

LOG LIKELIHOOD

$$l(\theta)=\log(L(\theta))=\log\left((2\pi\sigma^2)^{-n/2}\exp\left\{-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right\}\right)$$

$$=-\frac{n}{2}\log(2\pi)-\frac{n}{2}\log(\sigma^2)-\frac{1}{2\sigma^2}(y_i-\mu_i)^2$$

SCORE FUNCTION

$$l_*(\theta)=\begin{cases}a=\frac{d}{d\beta_1}l(\theta)\\ b=\frac{d}{d\beta_2}l(\theta)\\ c=\frac{d}{d\sigma^2}l(\theta)\end{cases}$$

$$a=\frac{d}{d\beta_1}\left(\cancel{-\frac{n}{2}\log(2\pi)}\cancel{-\frac{n}{2}\log(\sigma^2)}-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right)=\frac{d}{d\beta_1}\left(-\frac{1}{2\sigma^2}(y_i-\beta_1-\beta_2x_i)^2=-\frac{1}{2\sigma^2}\cdot2(y_i-\beta_1-\beta_2x_i)(-1)\right)$$

$$b=\frac{d}{d\beta_2}-\frac{1}{2\sigma^2}(y_i-\beta_1-\beta_2x_i)^2=-\frac{1}{2\sigma^2}\cdot2\cdot(y_i-\beta_1-\beta_2x_i)(-x_i)$$

$$c=\frac{d}{d\sigma^2}\left(\cancel{-\frac{n}{2}\log(2\pi)}-\frac{n}{2}\log(\sigma^2)-\frac{1}{2\sigma^2}(y_i-\mu_i)^2\right)$$

$$=-\frac{n}{2}\frac{1}{\sigma^2}-\frac{1}{2}(y_i-\mu_i)^2\left(\frac{-1}{(\sigma^2)^2}\right)$$

$$\frac{1}{x}=x^{-1}\to-\frac{1}{x^2}$$

- $\hat{\underline{\beta}}(\underline{Y})\dot{\sim}N_p(\underline{\beta};\sigma^2(X^TX)^{-1})$
- $\hat{\beta}_r(\underline{Y})\dot{\sim}N(\beta_r;(\sigma^2(X^TX)^{-1})_{(r,r)})$

BERNOULLI

$f(y_i;\theta)$

$$f(\pi_i)=(\pi_i)^{y_i}(1-\pi_i)^{1-y_i}\qquad \pi_i=\frac{e^{\tilde{x}_i^T\beta}}{1+e^{\tilde{x}_i^T\beta}}$$

LIKELIHOOD

$$L(\theta)=\prod\pi_i^{y_i}(1-\pi_i)^{1-y_i}$$

LOG LIKELIHOOD

$$l(\theta)=\log(L(\theta))=\log\left(\prod\pi_i^{y_i}(1-\pi_i)^{1-y_i}\right)$$

$$=\sum\log(\pi_i^{y_i}(1-\pi_i)^{1-y_i})$$

$$=\sum y_i\log(\pi_i)+(1-y_i)\log(1-\pi_i)$$

$$=\sum y_i\log\left(\frac{e^{\tilde{x}_i^T\beta}}{1+e^{\tilde{x}_i^T\beta}}\right)+(1-y_i)\log\left(\frac{1}{1+e^{\tilde{x}_i^T\beta}}\right)$$

$$=\sum y_i(\tilde{x}_i^T\beta)-\cancel{y_i\log(1+e^{\tilde{x}_i^T\beta})}-(1-\cancel{y_i})\log(1+e^{\tilde{x}_i^T\beta})$$

$$=\sum y_i\tilde{x}_i^T\beta-\log(1+e^{\tilde{x}_i^T\beta})$$

SCORE FUNCTION

$$l_*(\theta)=\begin{cases}a=\frac{d}{d\beta_1}l(\theta)\\ b=\frac{d}{d\beta_2}l(\theta)\end{cases}$$

$$a=\frac{d}{d\beta_1}\sum y_i\tilde{x}_i^T\beta-\log(1+e^{\tilde{x}_i^T\beta})=\frac{d}{d\beta_1}\sum y_i(\beta_1+\beta_2x_i)-\log(1+e^{\beta_1+\beta_2x_i})$$

$$=\sum y_i-\frac{e^{\beta_1+\beta_2x_i}}{1+e^{\beta_1+\beta_2x_i}}$$

$$b=\frac{d}{d\beta_2}\sum y_i(\beta_1+\beta_2x_i)-\log(1+e^{\beta_1+\beta_2x_i})$$

$$=\sum y_ix_i-\frac{e^{\beta_1+\beta_2x_i}}{1+e^{\beta_1+\beta_2x_i}}\cdot x_i$$

$$\hat{\underline{\beta}}(\underline{Y})\dot{\sim}N_p(\underline{\beta};J(\hat{\underline{\beta}})^{-1})$$

$$\hat{\beta}_r(\underline{Y})\dot{\sim}N(\beta_r;J(\hat{\underline{\beta}})^{-1}_{(r,r)})$$

POISSON

$$f(Y_i; y_i) = \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!} \qquad \mu_i = e^{\tilde{x}_i^T\beta}$$

LIKELIHOOD

$$L(\beta) = \prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}$$

LOGLIKELIHOOD

$$\ell(\underline{\beta}) = \log(L(\underline{\beta})) = \log\left(\prod \frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum \log\left(\frac{e^{-\mu_i}\mu_i^{y_i}}{y_i!}\right) = \sum -\mu_i + y_i \log(\mu_i) - \log(y_i!)$$

$$= \sum -\mu_i + y_i(\tilde{x}_i^T\beta) - \log(y_i!)$$

SCORE

$$\ell_*(\underline{\beta}) = \begin{cases} a = \frac{\partial}{\partial\beta_1}\ell(\underline{\beta}) \\ b = \frac{\partial}{\partial\beta_2}\ell(\beta) \end{cases}$$

$$a = \frac{d}{d\beta_1}\left(\sum -\mu_i + y_i(\tilde{x}_i^T\beta) - \log(y_i!)\right) = \frac{d}{d\beta_1}\left(\sum -e^{\beta_1+\beta_2 x_i} + y_i(\beta_1+\beta_2 x_i) - \log(y_i!)\right)$$

$$= \sum -e^{\beta_1+\beta_2 x_i} + y_i$$

$$b = \sum -e^{\beta_1+\beta_2 x_i}\cdot x_i + y_i x_i$$

$$\hat{\underline{\beta}}(\underline{Y}) \dot\sim N_p(\underline{\beta};\ J(\hat{\underline{\beta}})^{-1})$$

$$\hat{\beta}_r(\underline{Y}) \dot\sim N(\beta_r;\ J(\hat{\underline{\beta}})^{-1}_{(r,r)})$$